

Proposal Title:

Daily caffeine consumption and student self reported concentration during lecture

Research Questions:

- Is there a relationship between how much caffeine STA304H5 students consume each day and how well they can concentrate during lectures?
- Does this relationship still exist when we also consider students' sleep duration and stress levels?

Target Population:

Our target population is Fall 2026 STA304H5 students.

Sampling Method:

Since the participation is voluntary and we do not have a sampling frame for random selection, we will not use simple random sampling. Instead, our sampling method will be convenience sampling with voluntary participation. We will seek participation from both of the STA304 lecture sections.

Data Collection:: After receiving feedback on our proposal and making any needed changes, we will share a Google Forms questionnaire. The questions will ask about the past seven days. We will aim for at least 60 usable responses before the October 9 deadline. To encourage participation, we will:

Offer to complete other groups' surveys.

Send one reminder through the same permitted channels.

There will be eight substantive questions and once course specific ID questions for participation credit and to prevent duplicate response. We will keep only the latest response for each ID. It will be a voluntary participation and no ID will be shown in the public results.

Simple Statistical Test:

We will use simple linear regression for RQ1 to examine the relationship between average daily caffeinated beverage consumption and self-reported concentration during lectures. Average daily consumption will be calculated by adding the number of servings reported for coffee, tea,

energy drinks, and other caffeinated beverages (Q2-Q5). The concentration score from Q6 will be the response variable.

Advanced Statistical Test:

Multiple linear regression with model selection. For RQ2, we will examine the relationship while also considering sleep duration and stress. We will compare models that include beverage consumption alone, with sleep, with stress, and with both. This will assist us in deciding which combination of variables to include. The results will indicate associations, not causation of changes in concentration of variable to include. The results will indicate associations, not classification of changes in concentration by caffeine.

Questionnaire Link:

https://docs.google.com/forms/d/e/1FAIpQLSdgsWaZUTG8jXkPriXbxz9i1uvidPZ_-F0VTWR9CfN1h9XTlg/viewform?usp=publish-editor